

Walter Pairing Algorithms

Detailed flow charts for Swiss, round robin, and single elimination across 40-, 60-, and 100-card format families

Method: static analysis of executable Python code only. No external research and no Walter documentation was used. The diagrams describe what the current implementation does, including surprising or unsafe paths—not what tournament policy might intend.

Deck / format family	Swiss	Round robin	Single elimination
40-card Draft; Sealed branch exists	2-player; specialized R1 limited seating	2-player circle schedule; no limited R1 special	Random pairwise bracket
60-card Standard, Pioneer, Modern, Legacy, Vintage, Pauper; Old School behaves likewise	2-player random R1, standings/backtracking later	2-player circle schedule	Random pairwise bracket
100-card Commander	4-player random R1, standings pods later	Falls back to Swiss pods but retains RR round cap	Pairwise bracket path conflicts with Commander result model

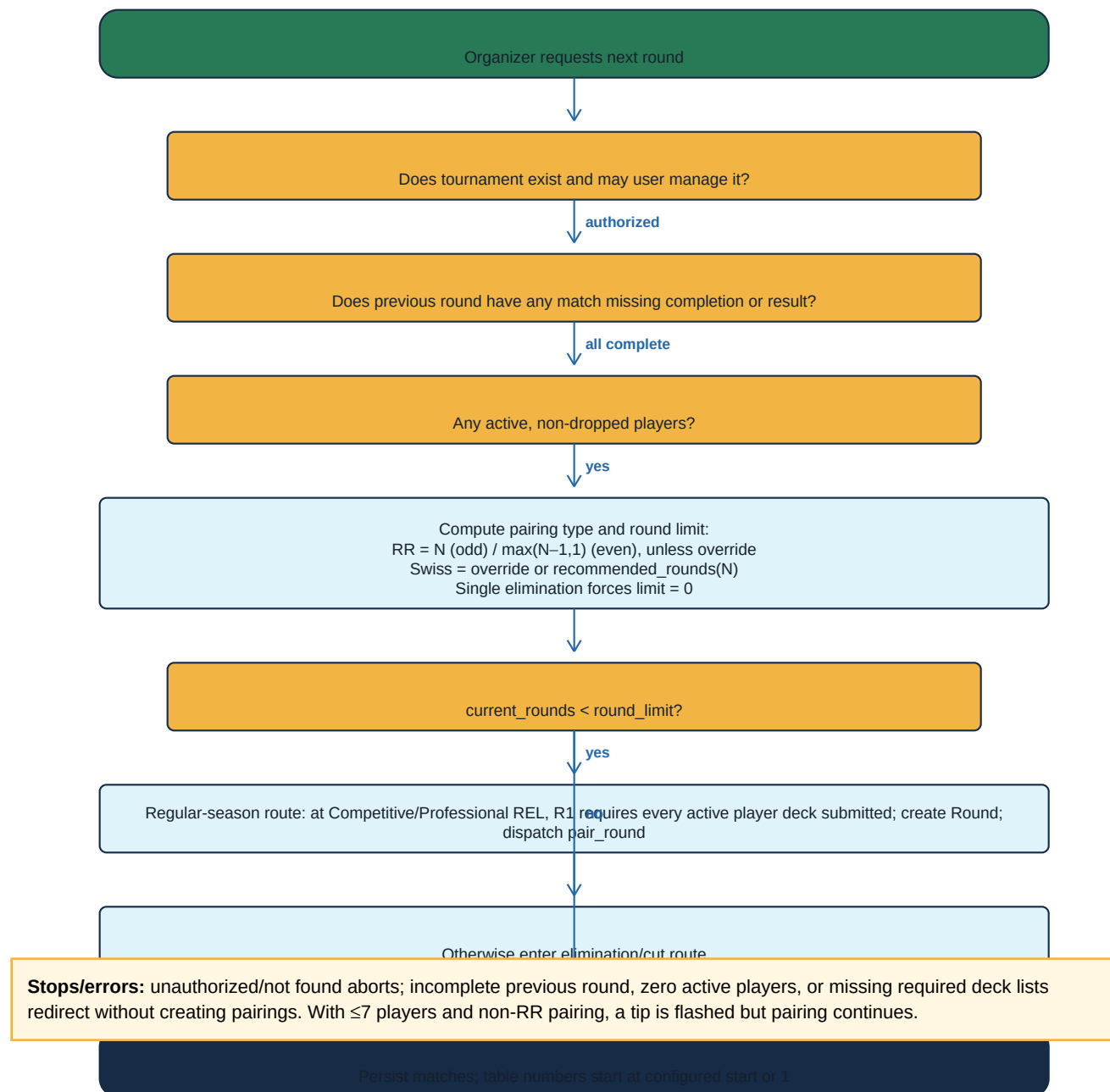
Primary code surfaces analyzed

- `app/pairing.py`: round limits, standings ordering, repeat detection, pod search, limited round one, round-robin rotation, and dispatch.
- `app/app.py`: format lists, the “pair next round” gate, round caps, cuts, and elimination advancement.
- `app/models.py`: tournament structure/pairing fields and two-player versus four-player match slots/results.

Generated 11 September 2026 (UTC) • Artifact intentionally created outside the Git repository.

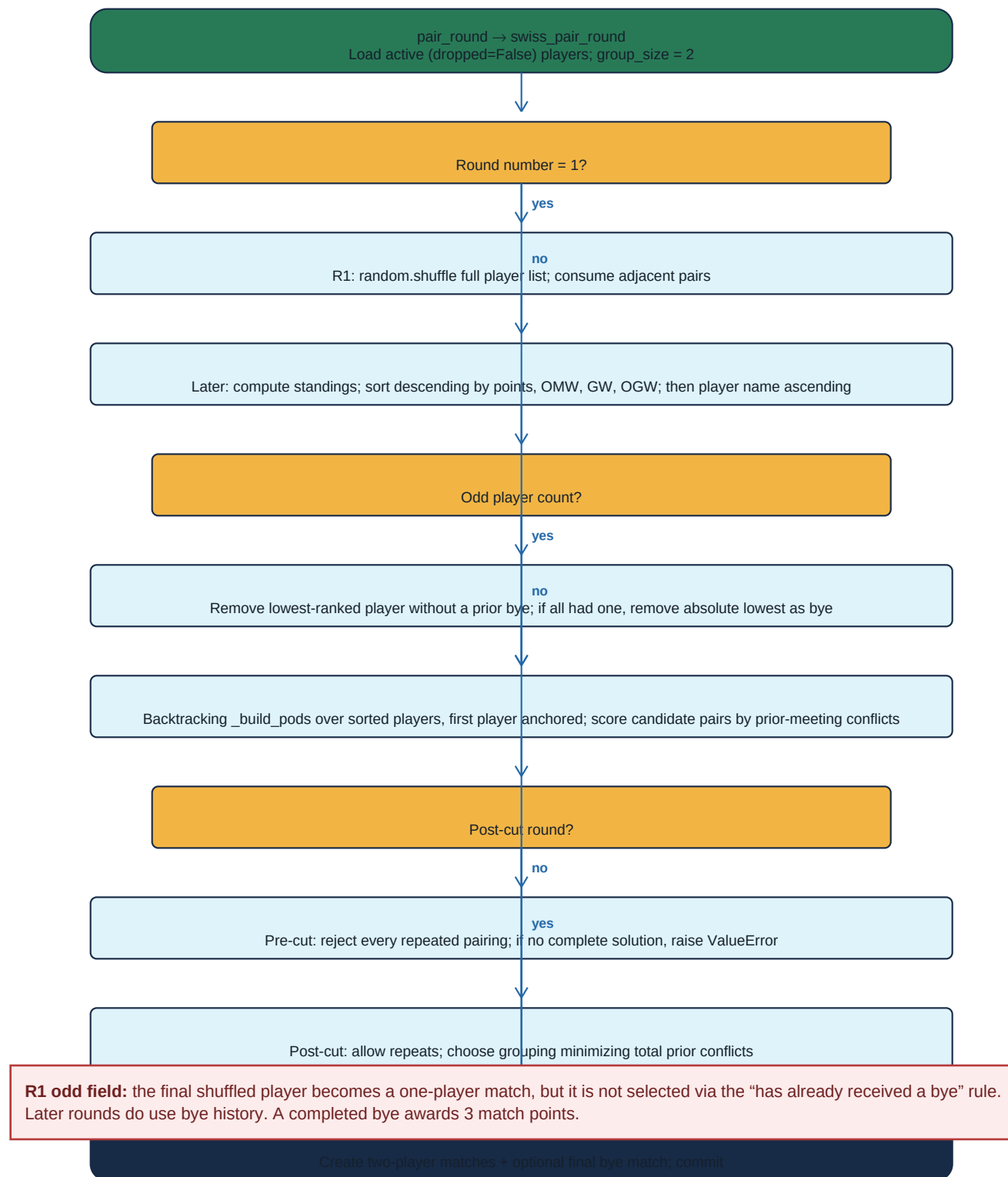
1. Shared “pair next round” gate

Runs before all three pairing families



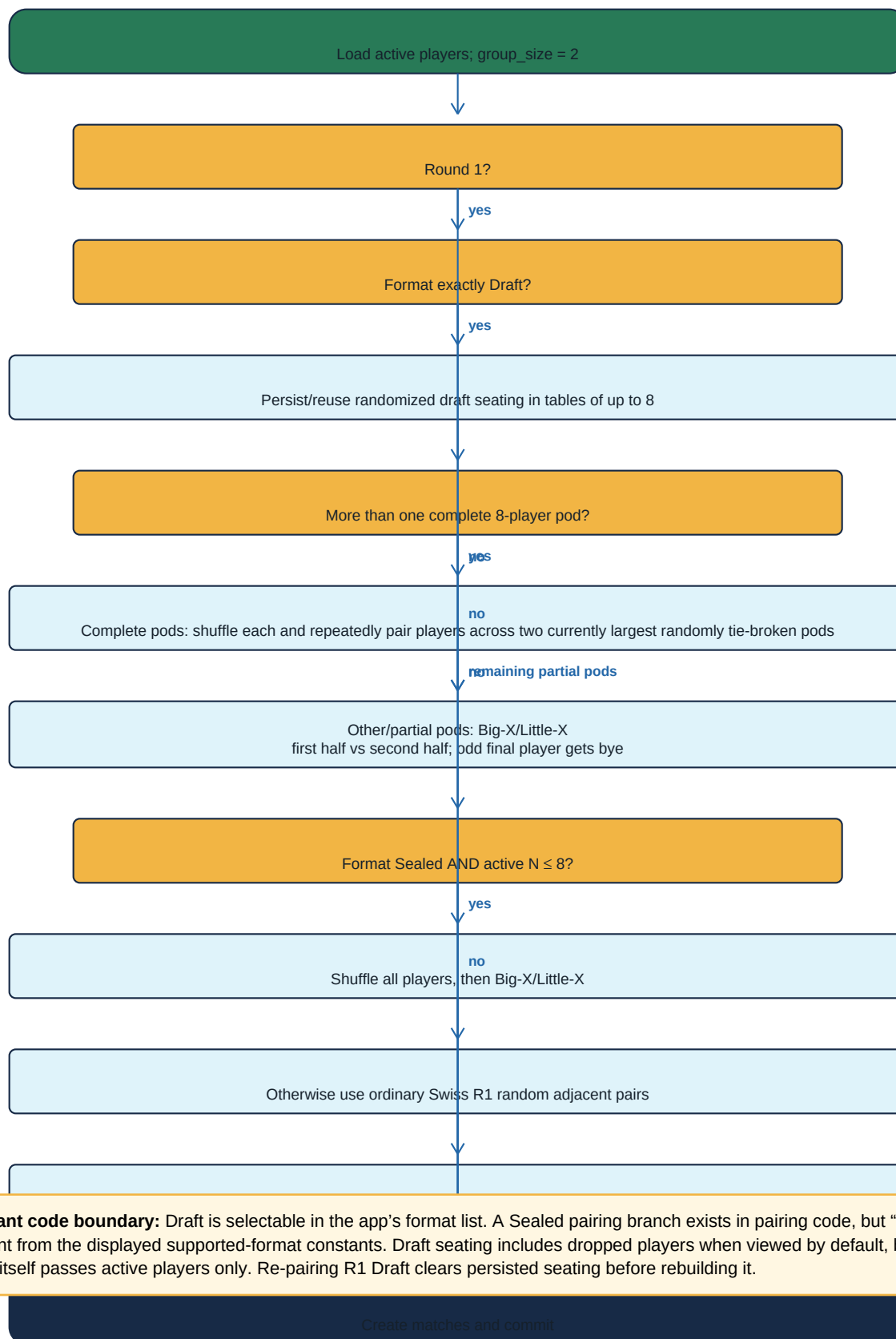
2. Swiss — 60-card family

Applies to all non-Commander, non-Draft/Sealed-special formats



3. Swiss — 40-card family

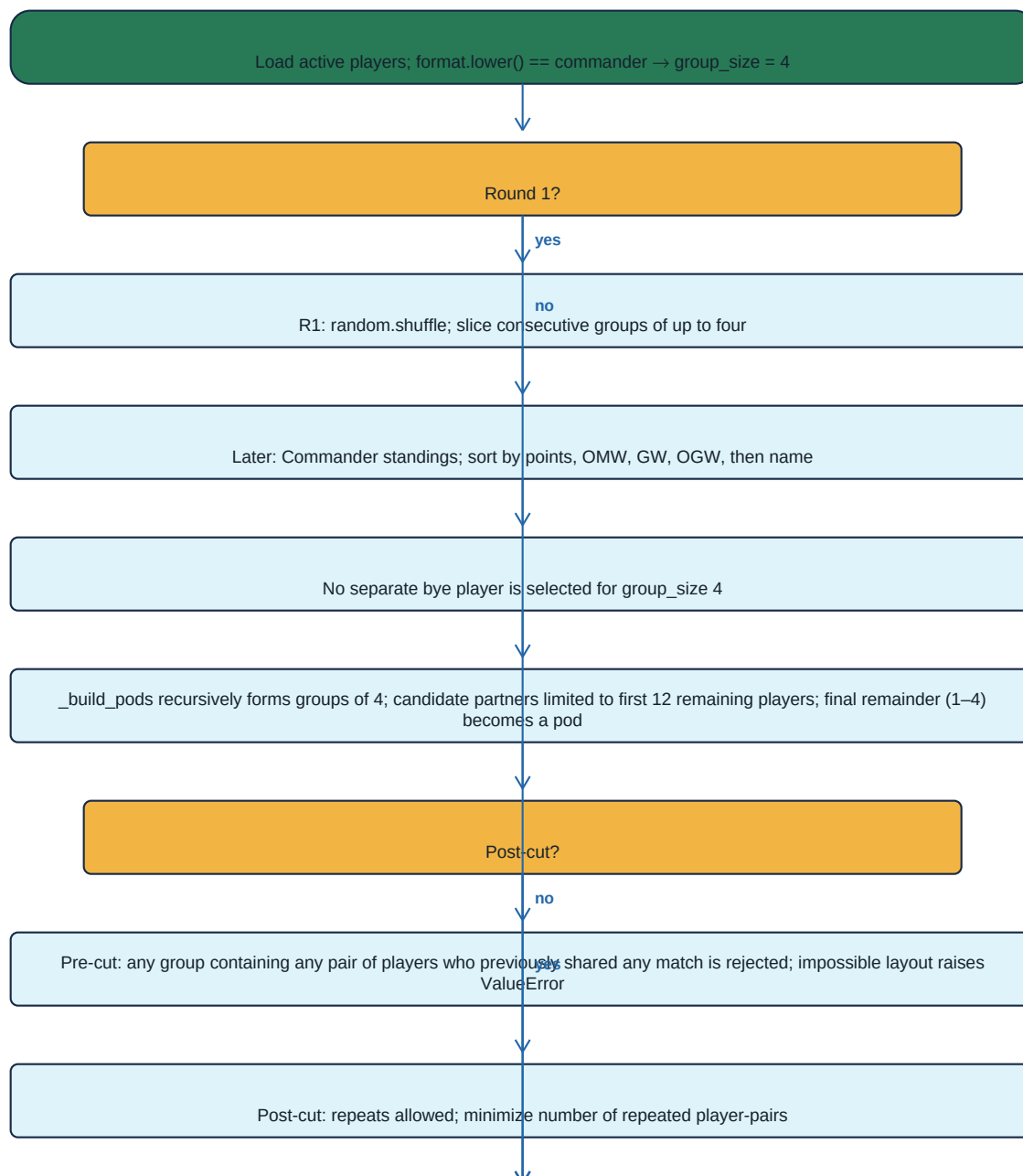
Draft and the implemented Sealed branch; pairings remain two-player



Important code boundary: Draft is selectable in the app's format list. A Sealed pairing branch exists in pairing code, but "Sealed" is absent from the displayed supported-format constants. Draft seating includes dropped players when viewed by default, but pairing itself passes active players only. Re-pairing R1 Draft clears persisted seating before rebuilding it.

4. Swiss — 100-card Commander

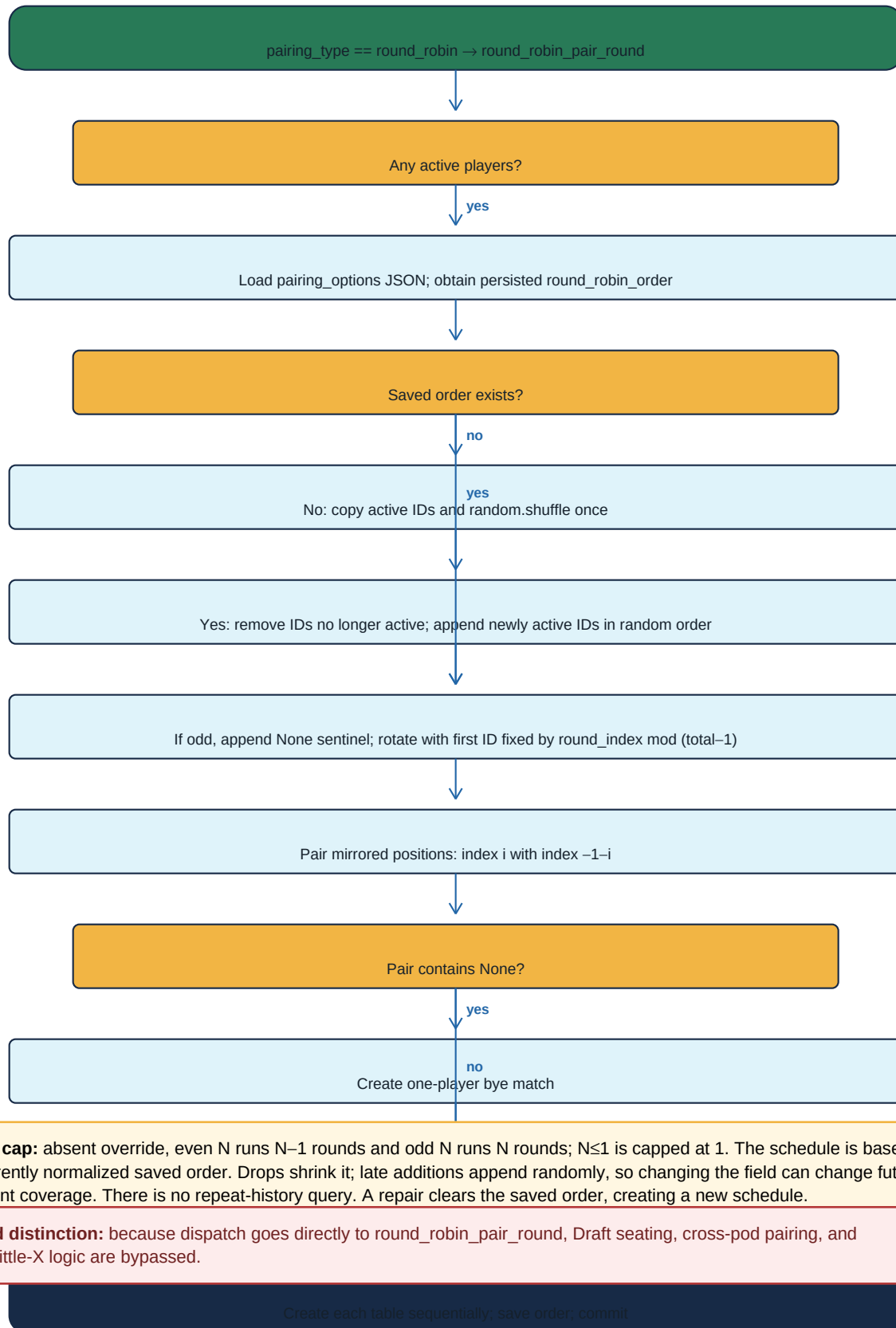
Four-player pod logic



Edge behavior: “have played” means co-appearance anywhere among four match slots. Thus every pair inside a prior pod conflicts. Unlike two-player Swiss, Commander does not remove one player for a bye; short pods are emitted as matches. Point configuration is parsed from comma-separated Commander values; placements 1–4 index that list, and a draw uses item 5 or defaults to 1.

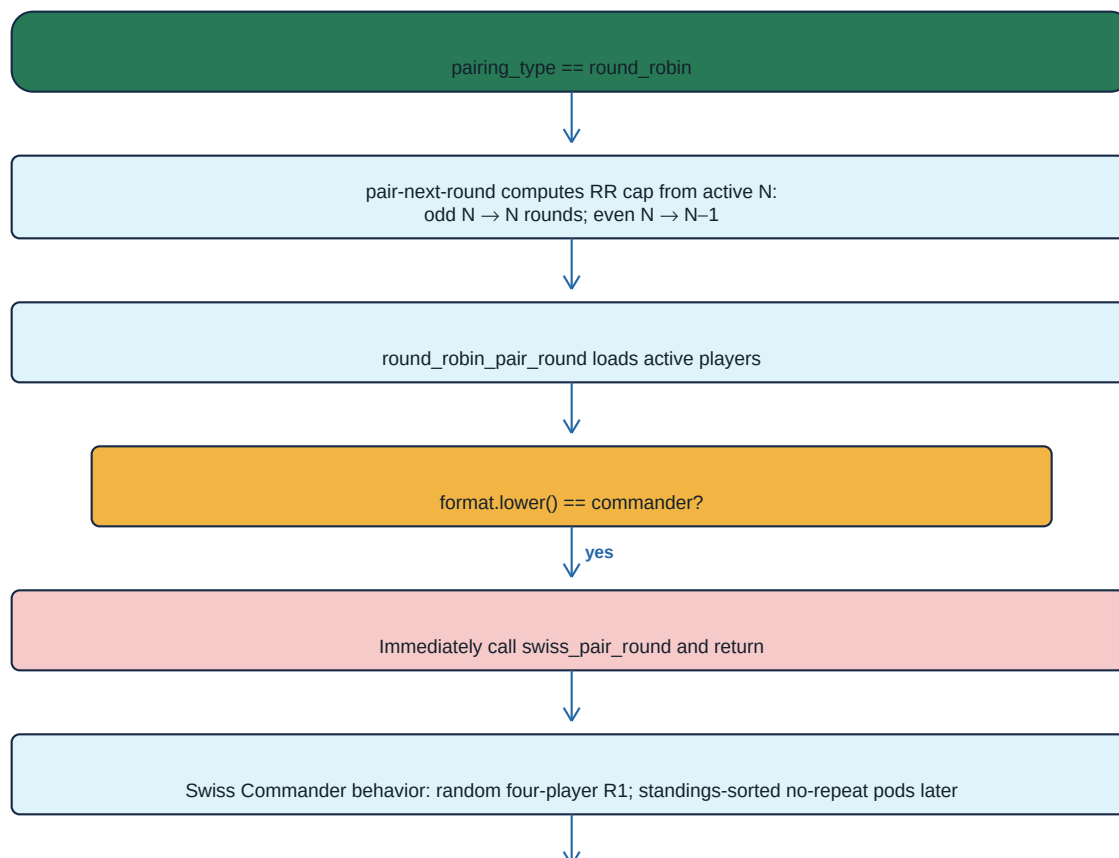
5. Round robin — 40-card and 60-card families

Same two-player circle-method implementation; no Draft/Sealed R1 specialty



6. Round robin — 100-card Commander

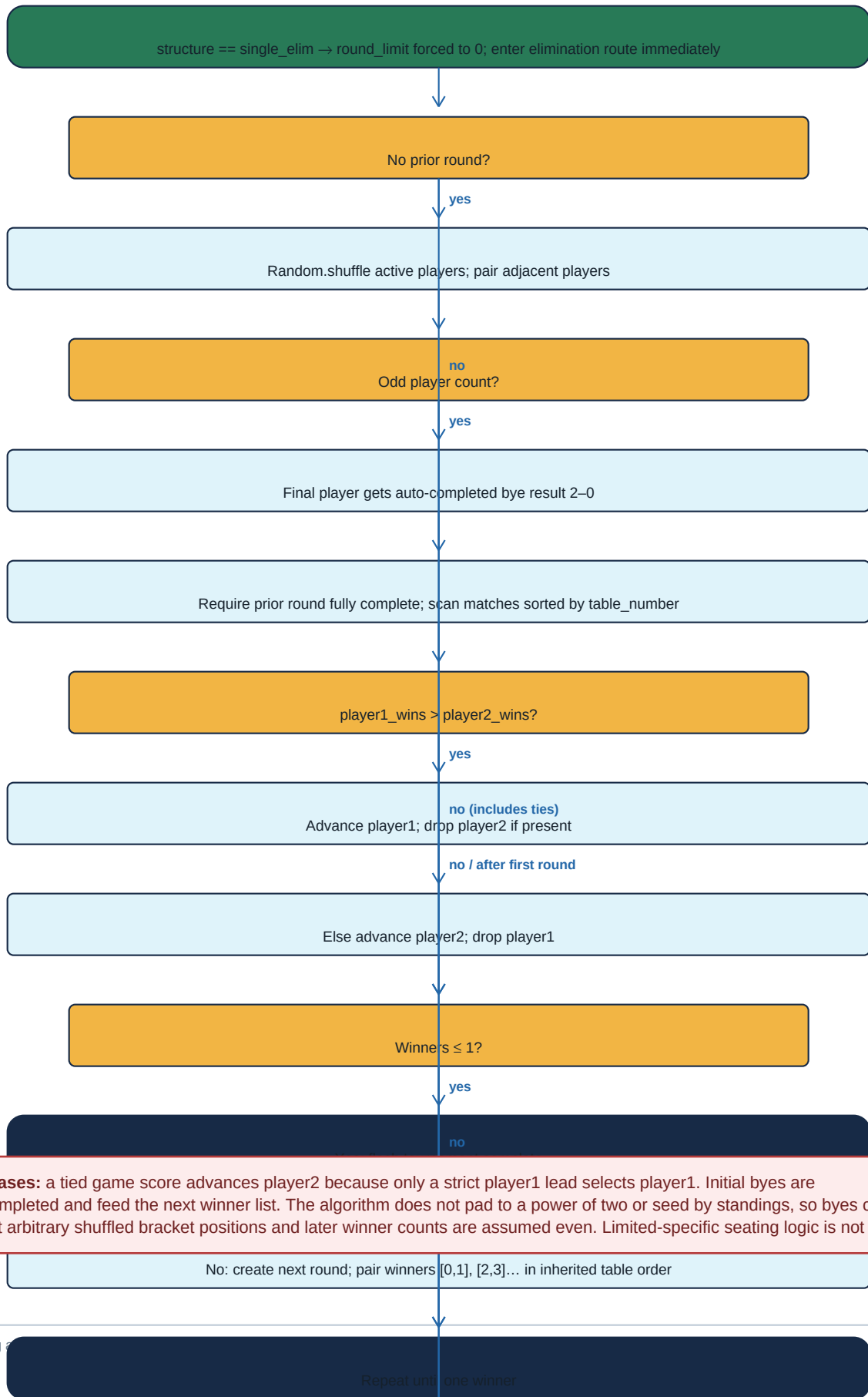
Requested type is not actually a Commander round robin



Consequence: the label and stopping rule are round robin, but the actual matching algorithm is Commander Swiss. It neither uses the circle rotation nor guarantees that every pair of players shares a pod. Pre-cut Swiss repeat avoidance may raise `ValueError` before the RR-derived number of rounds can be completed.

7. Single elimination — 40-card and 60-card families

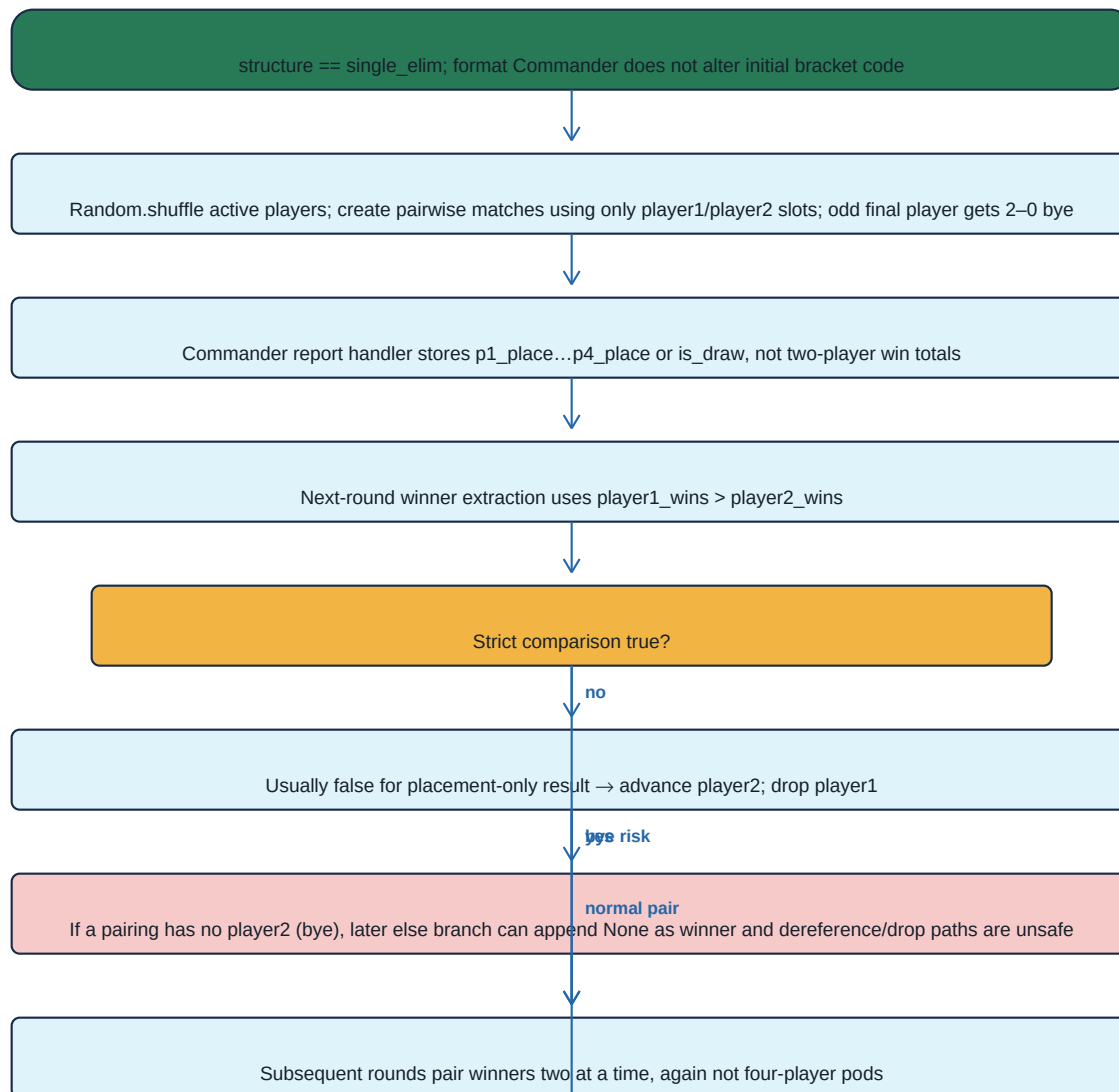
Random initial bracket; winners preserve table order



Edge cases: a tied game score advances player2 because only a strict player1 lead selects player1. Initial byes are auto-completed and feed the next winner list. The algorithm does not pad to a power of two or seed by standings, so byes can occur at arbitrary shuffled bracket positions and later winner counts are assumed even. Limited-specific seating logic is not used.

8. Single elimination — 100-card Commander

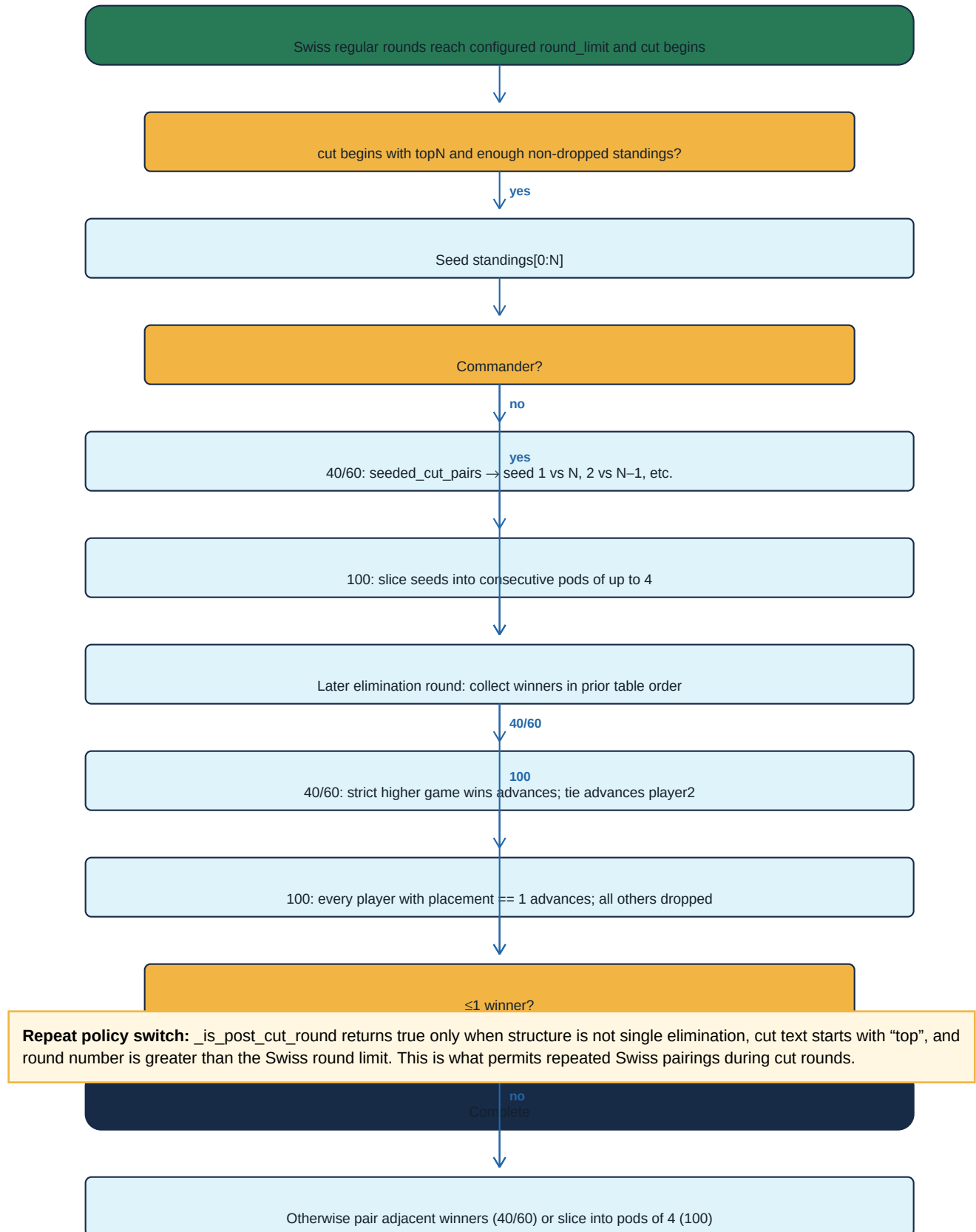
The current code path is internally inconsistent



Code-audit finding: although other elimination/cut code has Commander-specific "placement 1 advances" logic, the dedicated `structure == single_elim` branch does not. It creates pairwise matches and advances via two-player win fields. Treat this combination as unsupported/defective rather than assuming 4-player elimination semantics.

9. Swiss cut-to-elimination path

Distinct from selecting single elimination as the tournament structure



10. Edge-case register and operational implications

Observed directly from branches, indexing, persistence, and database queries

Case	Actual behavior / risk
Zero / one player	Zero active players is blocked by the route. One player: Swiss R1 emits a one-player match; RR schedules one bye round; single elimination emits an auto-completed bye, then needs a later request to declare completion.
Drops	Every pairing query excludes dropped players. RR removes dropped IDs from saved order. Swiss historical conflict checks still query old matches. Elimination explicitly drops losers.
Late entrants	Swiss sees any active player next round. RR appends missing IDs randomly to persisted order, altering the schedule's mathematical coverage.
Repeat pairings	Pre-cut Swiss requires a globally zero-conflict grouping and raises if none exists. Post-cut Swiss minimizes conflicts. RR trusts rotation and does not query history. Single elimination never checks repeats.
Odd / remainder fields	2-player Swiss and RR use one-player matches. Commander Swiss emits short pods. Single elimination only explicitly handles an odd player in the first round; later winner pairing indexes $i+1$ without a guard.
Randomness / repair	Swiss R1 shuffles; Draft seating persists; Sealed ≤ 8 reshuffles; RR saves order but randomizes slot orientation. Re-pair deletes unreported matches, clears RR order and Draft R1 seating as applicable, then reruns.
Malformed state	pairing_options invalid JSON silently becomes {}. RR rebuilds an order. Draft seating rebuilds from usable active IDs.
Tables	Every algorithm starts at start_table_number or 1 and increments once per Match, including byes.
Round override	Override replaces recommended Swiss rounds and RR natural schedule length. Single elimination always forces regular-round limit to 0.
Recommended Swiss caps	Code returns 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 rounds at player thresholds ≤ 8 / 16 / 32 / 64 / 128 / 226 / 409 / above 409.
Deck size itself	No pairing branch reads a numeric deck size. The 40/60/100 distinctions in this report map from format-name checks: Draft/Sealed, ordinary non-Commander formats, and Commander.

Bottom line: 40- and 60-card formats differ only in Swiss round one when the format string selects Draft or qualifying Sealed logic. Commander fundamentally changes Swiss to four-person pods. Commander round robin is a Swiss fallback, and Commander single elimination follows a contradictory two-player path.